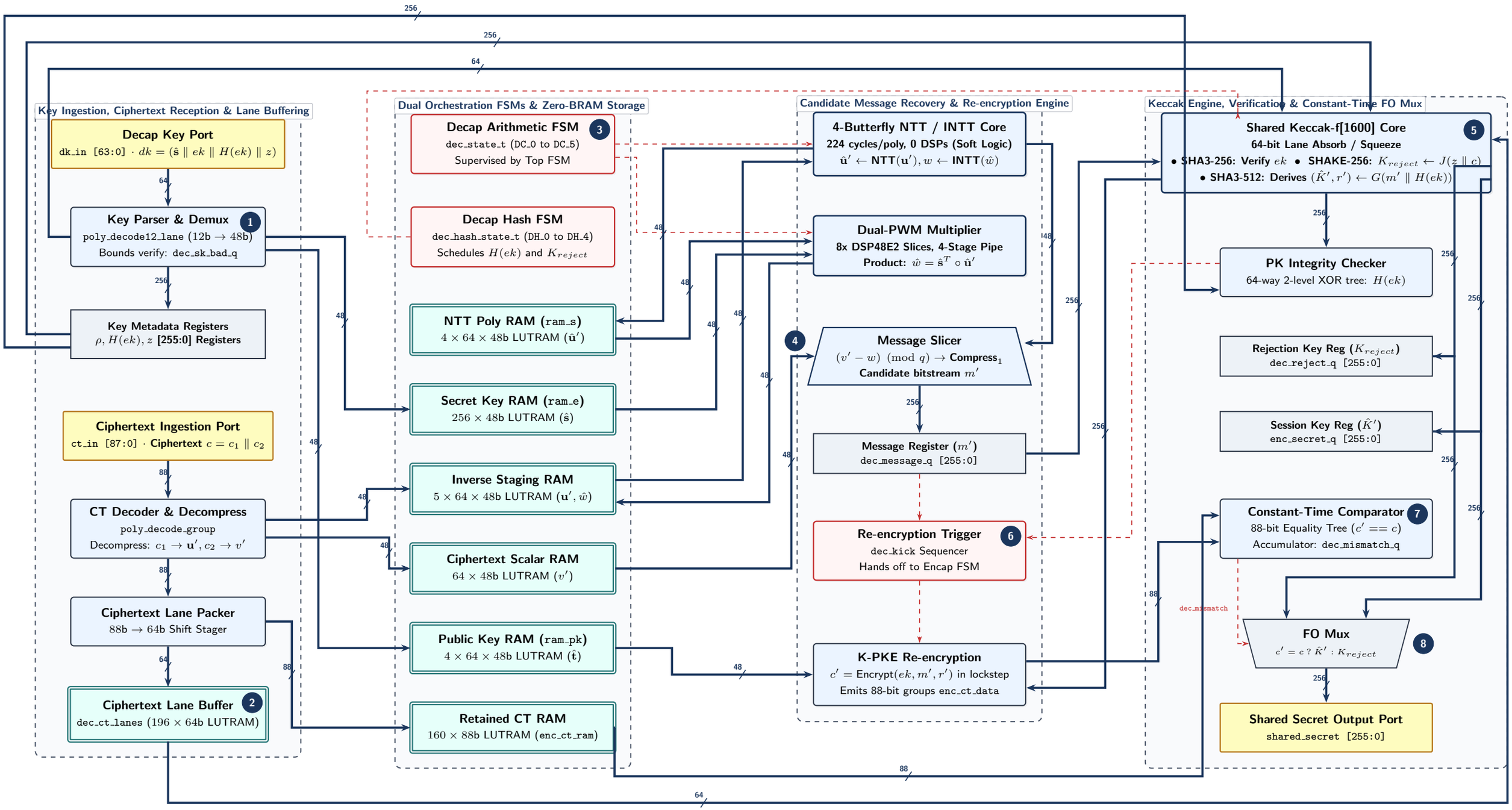
ML-KEM HPKA: Decapsulation Datapath & Constant-Time FO Transform Engine

Subsystem 03 Microarchitecture — RTL Modules: `mlkem_decap_datapath.svh` & `mlkem_keygen_top.sv`



ID	Component / Signal	Width / Format	Description
1	Decap Key Ingestion & Demux	64b Stream → 48b/256b	Parses $dk = (\hat{\mathbf{s}} \parallel ek \parallel H(ek) \parallel z)$, decodes secret vector $\hat{\mathbf{s}}$ into `ram_e`, stages public key ek into `ram_pk`, and registers seeds $\rho, H(ek), z$.
2	Ciphertext Buffer (`dec_ct_lanes`)	196 × 64b LUTRAM	Decompresses $c_1 \to \mathbf{u}', c_2 \to v'$ into staging RAMs and serializes 88-bit groups into 196 64-bit lanes for $J(z \parallel c)$ hash.
3	Dual Decap Orchestration FSMs	Control / Handshake	Arithmetic FSM (`DC_0`–`DC_5`) schedules NTT, dyadic multiplication, INTT, and slicing; Hash FSM (`DH_0`–`DH_4`) manages $H(ek)$ check and K_{reject}.
4	Candidate Message Recovery Slicer	48b Word → 256b Reg	Evaluates $w = \text{INTT}(\hat{\mathbf{s}}^T \circ \hat{\mathbf{u}}')$, subtracts $(v' - w) \pmod q$, and slices 1-bit coefficients via $\text{Compress}_1(v' - w)$ into 256-bit register m'.
5	Shared Keccak-f[1600] Sponge Core	64b Lane → 256b/512b	Executes SHA3-256 for ek integrity verification, SHAKE-256 for $K_{reject} = J(z \parallel c)$, and SHA3-512 for $(\hat{K}', r') = G(m' \parallel H(ek))$.
6	Re-encryption Trigger Sequencer	Handshake (`dec_kick`)	Asserts `dec_kick` upon message recovery and hash checks, transferring control to Encap FSM to regenerate $c' = \text{Encrypt}(ek, m', r')$.
7	Constant-Time Ciphertext Comparator	88b Parallel Lockstep	Compares regenerated c' against stored c group-by-group, accumulating differences into `dec_mismatch_q` in strictly invariant clock cycles.
8	Branchless Constant-Time FO Mux	256b Multiplexer	Emits valid session key $\hat{K}'$ if $c' == c$ or pseudo-random rejection key K_{reject} if $c' \neq c$, eliminating chosen-ciphertext timing side channels.